Raul Mendy

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EXPERIENCE

Software Development Engineer II | Volico October 2024 – Present

- Developed an AI-powered customer service agent using a RAG vector database and LLM with speech-to-text and text-to-speech capabilities, enabling automated resolution of common support requests following a 400-client contract expansion — reducing inbound calls to human agents by 60%.
- Designed and implemented a monitoring pipeline that collects data from 200+ building management devices across the data center via Modbus TCP and maps every data point into Zabbix for alerting.
- Replaced a deprecated AT&T email-to-SMS gateway with a self-hosted SMTP server that routed alerts by app and severity to the right on-call engineers, delivered via Twilio SMS and a Slack bot.

Software Development Engineer I | Florida Power & Light August 2023 – September 2024

- Led a team of 3 interns building PowerGPT, a conversational grid-analytics platform (React, Flask, PostgreSQL) that turned our IEEE-published research into tools engineers could query in plain language.
- Built ML services for fault classification across 46 fault types (TensorFlow/Keras), 24-hour probabilistic wind-power forecasting from National Weather Service data (LightGBM), and spaCy-based query parsing.

Research Associate | Florida Atlantic University May 2023 – August 2023

- Developed a Large-Scale Foundation (LSF) model and interface for multimodal data analysis, enhancing Prognostic Health Management of smart-grid power transformers and achieving 97% fault classification accuracy; published at IEEE ICMLA 2023.

SKILLS

Languages: Python, C++, C, JavaScript, Java, SQL

Backend: FastAPI, Flask, React, Node.js, Rails, PostgreSQL, Redis, Qdrant, Docker, Kubernetes

AI/ML: PyTorch, TensorFlow/Keras, LightGBM, spaCy, OpenCV, RAG, Vector Databases, NLP

Cloud & DevOps: AWS (EC2, S3), Google Cloud, Zabbix, GitLab CI/CD, Git, Postman

Protocols & Systems: Modbus TCP, SMTP, WebSockets, Twilio & Slack APIs, Real-Time Processing, Linux

PROJECTS

Smart Wildlife Camera System | FAU Senior Design for Skukuza Research Station, South Africa

- Engineered a solar-powered smart camera system using Raspberry Pi and OpenCV for real-time wildlife detection and image analysis in remote conservation areas.
- Worked with the networking team on a delay-tolerant (HDTN) design where rangers' vehicles collected images and analysis when in range; delivered the prototype to the station.

TrackFlights https://track-flights-web.vercel.app/

- Built an ETL pipeline that ingests live aircraft broadcasts from a Raspberry Pi ADS-B receiver (PiAware) near Miami International Airport into a cloud database.
- Deployed a live dashboard that displays arrivals, departures, delays, and airport performance metrics.

Encryption/Decryption Website https://mendyraul.github.io/pages/encryption.html

- Built a client-side tool to encrypt and decrypt text with DES or AES in ECB, CBC, and CFB modes; keys and text never leave the browser.

EDUCATION

Florida Atlantic University May 2023

B.S. Computer Science, Minor in Cybersecurity 3.8 GPA

- **Publication**: J. Accurso, **R. Mendy**, A. Torres, Y. Tang — **A ChatGPT-like Solution for Power Transformer Condition Monitoring** (IEEE ICMLA) https://ieeexplore.ieee.org/document/10459800

Tufts University May 2019

B.S. Community Health

LEADERSHIP

Captain, Argentina National Lacrosse Team | Buenos Aires, Argentina May 2018 – Present

- Played in the 2018 FIL World Lacrosse Championship in Netanya, Israel.

Defensive Coordinator, FAU Men's Lacrosse August 2022 – Present

- Design defensive game plans and coach player development for a roster of about 50 players each season.